

Research Plan for Doctoral Degree

Making Knowledge Available and Accessible through Artificial Intelligence

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1 Introduction

1.1 Research problems and questions

This research plan addresses the following research problems:

Problem 1: Institutions sit on tremendous amounts of tacit knowledge and acquired information, in large due to the digital revolution. However, most of this information is not used as it is scattered in unstructured or hard to interpret formats, like documents or human experience.

Problem 2: There is tremendous hype around Large Language Models (LLMs) currently and they are often seen as universal hammers. However, nobody really knows when these models should and should not be used.

We expect that answering the following research questions in a Ph.D. thesis will help solve the stated research problems:

Question 1: How can artificial intelligence (AI) and machine learning techniques be leveraged to make unstructured data usable?

Question 2: What are typical tasks and problems that are suitable for LLMs?

1.2 Research aim, objectives, and scope

The primary aim of this research is to explore how AI, LLMs, and different types of data storage and retrieval methods can be harnessed to make unstructured data usable, accessible, and beneficial for various stakeholders across different sectors.

Objective 1: Develop methodologies leveraging AI and machine learning to structure and index unstructured data from various sources, thus making it searchable and retrievable.

Objective 2: Identify and describe the types of tasks and problems suitable for LLMs, considering factors such as complexity, reliability, domain specificity, and data sensitivity.

Objective 3: Integrate and refine tacit knowledge management using AI-driven strategies to store and retrieve knowledge across various domains, thereby enhancing the discoverability, accessibility, and usability of tacit knowledge for various stakeholders.

This research will primarily focus on the application of AI and LLMs in processing and organizing unstructured data. While acknowledging that issues of knowledge accessibility can also be addressed through educational reforms, standards, policy changes, and improved documentation practices, this study will limit its scope to technological interventions using AI. The research will not cover the broader implications of AI deployment or compare AI methodologies against non-technological solutions in detail. The study will concentrate on English, Finnish and Swedish-language data sources.

2 State-of-the-art review

The recent rapid progress in the AI field have once again reignited faith in our ability to one day create machines exhibiting human-like cognitive capabilities [1]. Recent AI models, like LLMs, are beginning to excel at tasks that they were not originally trained for, and such models are now collectively being labelled as General purpose AI models [2]. The conversion of unstructured data to a structured format is one example of a task that LLMs were not originally trained for, but that the most recent models are capable of [3]. Recently development methods like retrieval augmented generation (RAG) is further making it possible for other LLMs to utilize the extracted information [4]. Consequently, there is a lot of hype in the air, but as recently pointed out in [5], nobody really knows what type of tasks that these AI systems will excel at.

3 Materials and methods

Data scientists and engineers are trained to analyze complex datasets to extract insights and patterns that can inform business operations. However, the data engineers' time is often spent retrieving, organizing, and pre-processing data, which are repetitive and time-consuming tasks. We argue that AI can be used as aiding tools for engineers and scientists to reduce time spent on repetitive data engineering tasks that can be automated. This way, the engineer's focus can be redirected toward areas initially intended for, i.e., validation, verification, and interpretation of results.

The reason we intend to use AI to addressing these issues as opposed to standardisation is that It is not possible to standardize existing text and there is too much variability how to interpret different layouts in unstructured data that one needs human-like cognitive capabilities to get the task done.

The introduction of Generic AI and Multimodal Large Language Models have opened up new ways to interact with and distill information, allowing for an exploratory approach to knowledge extraction [1] [6], as well as rapid prototyping and idea validation [7]. Vector databases [8], knowledge graphs [9] [10], and multilingual/multimodal LLMS combined with novel HCI approaches [11] allow for developing and evaluating novel knowledge extraction and retrieval methods. In this thesis, we propose applying AI technologies and digitization to make engineering knowledge more accessible and understandable.

The quantitative study will use secondary data from company partners in the projects. Examples of quantitative data we will use to demonstrate AI information extraction are engineering documentation, specifications, CAD models, and vessel operational data. Primary data on design, commissioning, and operations in the shipbuilding industry can be obtained through interviews and workshops. Data obtained from companies are primarily of a descriptive type, meaning that they are not obtained through planned experiments but represent everyday engineering work. If needed, experiment data from sea trials and simulations can also be used in the Virtual Sea Trial project.

4 Research environment

The research will be carried out in an international, collaborative research environment, consisting of consortia of research organizations and industry with expertise in maritime technology and simulation. The research is to be funded by the Co-Innovation project Virtual Sea Trial and an employee at the Research, Innovation, and Development department at Novia University of Applied Sciences. The main relevant research environments and consortia are listed below.

Novia Intelligent Systems Institute promotes collaboration with Finnish businesses by leveraging the collective expertise in AI, machine learning, and information management. Its team includes project experts, researchers from Novia UAS, and a group of teachers.

Virtual Sea Trial is a Business Finland-funded Co-Innovation research project for the period 1.1.2024–31.12.2026, consisting of four university partners and eight industry partners.

Nordic Cooperation on Maritime Autonomous Surface Ships (NorMASS) is a academic collaboration that explores new horizons in teaching, research, and simulation technology dedicated to Maritime Autonomous Surface Ships (MASS).

5 Timeline

5.1 Publication plan

The presented publication plan is in chronological order. The goal is to publish four peer-reviewed papers.

Paper I *Information retrieval from unstructured documents using large language models.* This paper aims to use large language models (LLMs) to extract and organize information from unstructured documents and make them searchable.

Paper II *Using LLMs to reduce bias in literature reviews.* Biases commonly surface in systematic literature reviews from the author using a limited number of sources, selected keywords when searching for information, and time constraints in the paper analysis. This study will examine whether LLMs can be used to streamline the scoping, clustering, keyword extraction, and summarizing processes for extensive collections of research papers within a given field to reduce review biases.

Paper III *Explaining tacit engineering knowledge.* Continuation of Paper I. Engineering practices often draws upon the experience and expertise of engineers and specialists. Language

barriers in engineering are not limited to the language spoken but also differ between fields of expertise and organizations. We propose to create an AI-driven context explaining steps to help by providing additional words that can make things clearer as you go along.

Paper IV *Transparent decision making in cities and municipalities.* Continuation of Paper I and III. Meeting protocols from cities and municipalities are public information, but the information is hidden in many ways as citizens rarely dig through the meeting protocols. This paper aims to utilize LLMs to convert unstructured meeting protocols into a structured, searchable knowledge graph.

5.2 Planned research exchange

The Virtual Sea Trial project provides funding for research exchanges. The following universities have already stated their intent to receive researchers: Royal Institute of Technology (KTH, Sweden), Tallinn University of Technology (Estonia), Reykjavik University (Iceland), and University of Southern Denmark (SDU). In addition to research exchange, Novia UAS encourages participating in international summer schools.

6 Discussion

In the past, there was not enough information available on many topics. However, today, information exists in abundance, but it is often just not available. This thesis project seeks to address this very general problem and thus benefit humans broadly across multiple sectors of society. Everyone in society stands to gain from organizing and making abundant information accessible. The methods to be developed in this project are versatile and can be implemented in various contexts, extending beyond engineering and academia.

References

- [1] S. Bubeck, V. Chandrasekaran, R. Eldan, J. Gehrke, E. Horvitz, E. Kamar, P. Lee, Y. T. Lee, Y. Li, S. Lundberg, *et al.*, “Sparks of artificial general intelligence: Early experiments with gpt-4,” *arXiv preprint arXiv:2303.12712*, 2023.
- [2] T. Madiega, “General-purpose artificial intelligence.”
- [3] J. Dagdelen, A. Dunn, S. Lee, N. Walker, A. S. Rosen, G. Ceder, K. A. Persson, and A. Jain, “Structured information extraction from scientific text with large language models,” *Nature Communications*, vol. 15, no. 1, p. 1418, 2024.
- [4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, *et al.*, “Retrieval-augmented generation for knowledge-intensive nlp tasks,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [5] C. Fraser, “Generative ai is a hammer, and no one knows what is and isn’t a nail,” *Medium*, 2024.
- [6] H. S. Zheng, S. Mishra, X. Chen, H.-T. Cheng, E. H. Chi, Q. V. Le, and D. Zhou, “Take a step back: Evoking reasoning via abstraction in large language models,” 2024.

- [7] I. Arawjo, C. Swoopes, P. Vaithilingam, M. Wattenberg, and E. Glassman, “Chainforge: A visual toolkit for prompt engineering and llm hypothesis testing,” 2023.
- [8] Z. Jing, Y. Su, Y. Han, B. Yuan, C. Liu, H. Xu, and K. Chen, “When large language models meet vector databases: A survey,” *arXiv preprint arXiv:2402.01763*, 2024.
- [9] L. Yang, H. Chen, Z. Li, X. Ding, and X. Wu, “Chatgpt is not enough: Enhancing large language models with knowledge graphs for fact-aware language modeling,” *arXiv preprint arXiv:2306.11489*, 2023.
- [10] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, “Unifying large language models and knowledge graphs: A roadmap,” *IEEE Transactions on Knowledge and Data Engineering*, 2024.
- [11] E. Bozkir, S. Özdel, K. H. C. Lau, M. Wang, H. Gao, and E. Kasneci, “Embedding large language models into extended reality: Opportunities and challenges for inclusion, engagement, and privacy,” 2024.